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Live Streaming Gap Report (6 Jul 2026).

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I've reviewed the **Live Streaming Gap Report (6 Jul 2026)**.

This is the first YohPal module that I would **not** approve for production in its current state.

Unlike the main YohPal Live application, whose recent reports show it approaching production readiness, the **Live Streaming module still contains four unresolved blocking defects in the Flutter client**, even though the server-side stack (mediasoup SFU, signaling, Docker, TURN, JWT authentication, tests, Android foreground service, and DevOps scripts) is reported as complete.

Executive Assessment

Server Side

The report indicates the backend is production-ready:

-  mediasoup SFU
-  Signaling handlers
-  JWT authentication
-  Docker/TURN stack
-  Smoke tests
-  Android foreground service
-  DevOps scripts

No further server architecture work is required at this stage.

Flutter Client

This is where the remaining blockers exist.

The report identifies:

- **4 Blocking issues**
- **1 Medium issue**
- **1 New Medium finding**

and concludes:

"Module not ready for integration or testing."

Priority Analysis

BLOCKER 1

Missing

`flutter/pubspec.yaml`

Without this:

- module cannot compile
- package cannot resolve
- CI cannot build

This must be fixed first.

BLOCKER 2

Infinite recursion

`YohPalPeerFactory.createPeerConnection()`

The factory method recursively calls itself, resulting in a `StackOverflow` every time a peer connection is created. This affects both

broadcaster and viewer controllers.

BLOCKER 3

DTLS echo bug

This is the most serious streaming issue.

Instead of sending the client’s DTLS fingerprint, the Flutter client echoes the server’s fingerprint back during `connectTransport`, preventing the DTLS handshake from completing.

The result is:

- no media flow
- viewers receive no video
- broadcasters cannot stream

The report estimates this fix at approximately **2 hours**.

BLOCKER 4

Deprecated

`Color.withOpacity()`

Nine occurrences remain.

Current impact:

- build warnings

Future impact:

- Flutter 4 build failure

These should be replaced with:

`withValues(alpha: ...)`

Medium

Signaling requests currently have no timeout.

If the server never replies:

the UI can remain loading indefinitely.

Adding a timeout prevents controllers from hanging.

New Finding

`YohPalRemoteMediaAttacher` is currently unused.

The report recommends either:

- wiring it into the viewer consume flow, or
 - removing it after the DTLS fix is completed.
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My Recommendation

I would **not integrate Live Streaming into the production application yet**.

I would first complete a dedicated stabilization sprint focused only on Live Streaming.

This avoids introducing instability into the otherwise production-ready YohPal Live application.

Revised Readiness

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Component	Readiness
Main YohPal Live App	99–100%
Theme System	98%
Zero-Wait Feed	99%
Wallet	100%
YCIOS	98%
Live Streaming Flutter Client	≈75%
Live Streaming Server	100%

The Live Streaming Flutter client is now the primary engineering blocker.

NEXT SMART MOVE

Build: YOHPAL LIVE STREAMING PHASE LS-1 – CLIENT STABILIZATION SPRINT

Scope:

- Sprint A – Build Foundation
- Create flutter/pubspec.yaml
 - Verify standalone module build
 - Verify dependency resolution

- Sprint B – RTC Stability
- Rename YohPalPeerFactory.createPeerConnection() to buildPeerConnection()
 - Update all broadcaster/viewer call sites
 - Eliminate recursion

- Sprint C – DTLS Transport Fix
- Generate client SDP
 - Extract client fingerprint
 - Send client DTLS parameters
 - Validate media flow
 - Verify broadcaster → viewer stream

- Sprint D – Flutter Modernization
- Replace all Color.withOpacity() with withValues(alpha:)
 - Remove Flutter 4 deprecation warnings

- Sprint E – Signaling Stability
- Add timeout(Duration(seconds:10)) to signaling requests
 - Display retry UI
 - Prevent indefinite loading

- Sprint F – Remote Media
- Integrate YohPalRemoteMediaAttacher into viewer consume flow
 - Remove dead code
 - Validate remote renderer

- Sprint G – Certification
- flutter analyze
 - flutter test
 - Android streaming test
 - iPhone streaming test
 - 30-minute continuous live stream
 - Multiple viewers

- Reconnection tests
- Background/foreground transitions

Deliverables:

- Live Streaming Stabilization Report
- Streaming Performance Report
- WebRTC Certification Report
- DTLS Validation Report
- Android/iOS Streaming Benchmark
- Live Streaming Production Certificate

Success Criteria:

- ✅ Standalone module builds successfully
- ✅ No recursion
- ✅ DTLS handshake succeeds
- ✅ Media flows correctly
- ✅ No Flutter 4 deprecation warnings
- ✅ Signaling cannot hang indefinitely
- ✅ Remote media attaches correctly
- ✅ Android and iOS streaming validated
- ✅ Ready for integration into YohPal Live

Outcome:

The Live Streaming module becomes production-ready and can then be integrated into the already mature YohPal Live platform without introducing instability.